Controller Supplement

Icon	Status	Functions or meanings	Remark
	Lights out	Currently in off or non-hot water mode	Display on/off status
	Constant flashing	Currently in hot water mode on	Display on/off status
	Lights out	Currently in off or non-heating mode	Display on/off status
	Constant flashing	Currently in heating mode	Display on/off status
	Lights out	Currently in off or non-cooling mode	Display on/off status
	Constant flashing	Currently in cooling mode	Display on/off status
	Lights out	Currently in off or non-floor heating mode	Display on/off status
	Constant flashing	Currently in floor heating mode	Display on/off status
	Constant flashing	Silent mode / Night mode	Power on display
	Constant flashing	Powerful mode	Power on display
	Constant flashing	Intelligent Mode	Power on display
	Constant flashing	Electric auxiliary heating work (AC, hot water electric heating)	Power on display
	1s flashing	Electric auxiliary heat quick heat mode is enabled	Power on display
	2s flashing	Electric auxiliary heat sterilization mode is enabled	Power on display
	Flashing	WIFI Allocation	
	Constant flashing	WIFI connection successful	
IN	Constant flashing	Represents water intake	
OUT	Constant flashing	Represents water discharge	
RT	Constant flashing	Represents actual temperature/room temperature	
SET	Constant flashing	Representative Settings	
°C	Constant flashing	Display Celsius temperature	
°F	Constant flashing	Display Fahrenheit temperature	
%	Constant flashing	Show percentage	
88.8	Constant flashing	Display of actual values, set values and fault codes	
	Flashing	Circulating water pump: freeze-proof operation	
	Constant flashing	Circulating water pump: normal operation	
	Constant flashing	Hydration valve open	
	Constant flashing	Return water valve open	
	1Hz flashing	Activate timed water return feature	
	2Hz flashing	Activate manual water return function	
	Constant flashing	High, medium and low water levels are displayed	
	Constant flashing	The water supply valve	

Icon	Status	Functions or meanings	Remark
	Constant flashing	Photovoltaic mode/solar heating	
	1 Hz flashing	Start PV timing	
	Flashing	Currently in shutdown and refrigerant recovery state	
	Constant flashing	Currently on and defrosting	
	Constant flashing	Enter maintenance status	
	Constant flashing	Alarm is currently occurring	
	Constant flashing	Current button is locked	
	Constant flashing	Compressor operation	
	Constant flashing	High wind operation of the fan	
	Constant flashing	Fan low wind operation	
	1 second flash	Ventilation mode: high speed wind	
	2 second flash	Ventilation mode: low speed wind	
	Constant flashing	Online networking	
	Constant flashing	Display the current network unit number	
88.8	Display	Display real-time time	
	Always bright	Enable timed work mode	
ON	Display	Currently in the power-on timer period	
ON	Flashing	Current set working period start time	
OFF	Display	Currently in power-on timed non-working hours	
OFF	Flashing	Current set working period end time	
123	Flashing / Light out	Timed working hours 1, 2, 3, always on when set or when the clock enters this period, off in the rest of the case	
	Display	Show current week 1, 2, 3, 4, 5, 6, 7	

2. Controller Manual

2.1 **On/off button “ ”:** when the power is off and unlocked, press and hold the key “ ” for 1S, the unit will start; when the power is on and unlocked, press and hold the " " key, the unit will stop running; if the line controller is in the "lock key If the line controller is in the locked state, press and hold the " " key for more than 3 seconds to release the lock.

2.2 **Mode key “ ”**: When the line controller is on and unlocked, press the " " key to select the operation mode of the unit (different modes can be selected according to different models: floor heating, cooling, heating, hot water ..... etc.).

2.3 **Temperature Setting**

2.3.1 **For Heating/Cooling/DHW**

2.3.1.1 (When there is no curve setting) when the line controller is on and unlocked, press "△" or "▽" key, then press "△" or "▽ "△" or "▽" key to adjust the current mode set temperature; press and hold the "△" or "▽" key for more than 0.5 seconds to quickly increase or decrease, stop operation after 5S or press the on/off key to return to normal display; combination mode (for example: heating + Hot water), press the " " key to switch the temperature setting of another mode, the icon of the corresponding mode flashes at 1Hz when the temperature is set.

2.3.1.2 (When there is curve setting) when the line controller is on and unlocked, press "△" or "▽" key to display the current set temperature curve, then the curve setting value flashes, then press "△" or "▽" key can switch different curve control, combination mode (for example: heating + hot water), press " " key to switch the curve display and setting of another mode. After stopping the operation 5S or press the“ ” key to return to normal display.

2.3.1.3 **Curve control setting:** Press "△" or "▽" to enter temperature setting in the state without curve setting, press " " for 5S to enter curve setting, the display shows OFF to cancel curve control, press "△" or "▽" to select the corresponding curve control at this time: press "△" or "▽" to enter curve selection in the state with curve control, the curve control can be canceled by setting the curve to OFF.

2.3.1.4

Refrigeration curve representation: CH1, 2, 3, 4, 5, 6, 7,8, represents the cooling high temperature curve 1, 2, 3, 4, 5, 6, 7,8

Refrigeration curve representation: CL1, 2, 3, 4, 5, 6, 7,8, represents refrigeration low temperature curve 1, 2, 3, 4, 5, 6, 7,8,

The heating curve is represented by: HH1, 2, 3, 4, 5, 6, 7,8, represents the heating high temperature curve 1, 2, 3, 4, 5, 6, 7,8,

Heating curve: HL1, 2, 3, 4, 5, 6, 7,8, The heating curve represents the low temperature curve 1, 2, 3, 4, 5, 6, 7,8,

Floor heating curve: GH1, 2, 3, 4, 5, 6, 7,8, The floor heating curve represents the high temperature curve 1, 2, 3, 4, 5, 6, 7,8,

Floor heating curve: GL1, 2, 3, 4, 5, 6, 7,8, representing the floor heating low temperature curve 1, 2, 3, 4, 5, 6, 7,8,

Hot water curve: H1, 2, 3, 4, 5, 6, 7,8, on behalf of the hot water curve 1, 2, 3, 4, 5, 6, 7,8.

2.4 **Quick heat function:** When the power is on, in non-cooling mode, press and hold " " + "△" for 3s to enter/exit the quick heat function.

LCD display: when the electric heating is on, the " " "symbol lights up, when the electric heating is not on " " "1Hz blinks to show the execution of the switch operation to turn off the quick heat function.

2.5 **Forced defrost function:** In the power-on state, in non-cooling mode, long press "M" + "▽" for 5s to enter the forced frosting. LCD display: the frosting icon " " lights up.

2.6 **IPLV test mode:**

For experimental test of inverter:

At power on, press and hold "On/Off" + "Timer" + " Ⓜ " for 5s to enter; press " ⏻ " to exit.

LCD display: IPLU is displayed in the timing area.

2.7 **Refrigerant recovery function:**

In the off state, within 5min of power on, long press " Ⓜ " + "▽" for 10s to enter the refrigerant recovery; press the " ⏻ " key to exit.

LCD display: frosting icon " ❄ " 1Hz flashing.

Inverter two-tier and three-tier temperature zone flashing to show the low pressure temperature value.

2.8 **Water line evacuation mode/forced on pump:**

In the off state, long press " ⏻ " + "△" + "▽" for 5s to enter; press again or directly press " ⏻ " key to exit. LCD display: pump icon " 🚰 " flashing

2.9 **Operation parameters query:**

2.9.1 Enter to view: long press "▽" 3S under the regular interface to enter the operating parameters query, enter the parameter query state, the temperature display area shows the parameter serial number, the timing area shows the parameter content. Forced operation parameters table differs according to the model, refer to the attached table or motherboard function manual.

3.9.2 Check operation and exit: After entering parameter query, press "△" or "▽" key to scroll through the "operating parameters"; press " ⏻ " key or 60 seconds without key operation automatically exit to view the status of parameters.

2.10 **Parameter Setting**

2.10.1 When the main interface is displayed, press and hold " Ⓜ " for 5S to enter the parameter query, in the query state, the parameter serial number flashes; press " Ⓜ " to enter the parameter setting state, the parameter value starts to flash, at this time, press "△ "or "▽" key to modify the parameters, press " Ⓜ " to determine the value of the parameter modification. Press the " ⏻ " key or 60 seconds without key operation to exit the parameter state automatically, without saving the modified parameter value.

2.10.2 There are two levels of parameters that can be set for two or three inverters, press and hold " Ⓜ " for 5S to switch the parameter query.

2.10.3 Parameter table varies depending on the model, refer to the attached table or motherboard function manual

2.11 **Display faults:**

2.11.1 When there is a fault in the unit, the fault is displayed in the timing area and the fault code is displayed cyclically while " 🔔 " flashes, and the normal display is restored when the fault is eliminated. The description of the fault code is shown in “Trouble Analysis”.

2.12 **Clock Settings:**

2.12.1 **Enter the clock setting:** the line controller unlocked state, press the " 🕒 " key, the clock area flashing display, such as the start of the week function” 8 ” will flash together, indicating that enter the clock setting state;.

2.12.2 **Clock setting operation:** Enter the clock setting state, press the " 🕒 " key to switch to set the week, hour and minute settings, the set value flashes at this time, press the "△" or "▽" key to modify the corresponding value, after setting After setting the minute value, press the "Timer" key or " ⏻ " key again or 5 seconds without key operation to save the current setting and exit automatically.

In the clock setting state, press and hold the " 🕒 " key for 3 seconds to turn on/off the weekly function; when the weekly timer is enabled, enter the clock setting first, then press the " 🕒 " key to enter the hour setting after setting, and skip the weekly setting directly when the weekly function is turned off.

2.13 **Set on/off timer control**

2.13.1 Timer control has 3 groups of timer, 1~3 groups, each group can be set as "timer start" and "timer stop"; default is "invalid" state, i.e. the timer start and timer stop are the same time.

2.13.2 To access the timing settings:

2.13.2.1: Long press the " " key for 3 seconds under the main interface to enable or disable the timer working mode.

2.13.2.2: When the day starts timing, all timing of the day is cancelled; when the day does not start stopping, the timing setting is entered.

2.13.2.3: When the week function is activated, after entering the timing setting, first enter the week selection, then "" "flash, press the "△" or "▽" key to select the week that needs to set the timing, and then proceed to the next operation. Skip this step when week function not start.

2.13.2.4 Short press the " " key, "1", "ON" icon at the bottom left of the screen, the hour display area flashes, it means enter the "first group of timing Start" setting state, digital flashing at hour, press "△" or "▽" key to modify the time, then press "Timer" key to confirm the modification and go to the minute Setting, digital flashing at the minute, press "△" or "▽" to modify the time, then press "Timer" to confirm the modification; at the same time, enter "1 ", "OFF" icon display, setting status, digital flashing display at hour, press "△" or "▽" key to modify the time, and then press "Timer" key to confirm the modification and go to the minute setting, digital flashing display at the minute, then press "△" or "▽" key to modify the time, then press " " "key to confirm the modification and start "Timing of time slot 1", and enter the setting of time period 2.

2.13.2.5 The operation of "Time slot 2 and 3" settings is the same as above.

2.13.3 **Exit setting timing:** When the timer is set, press the " " key or no key operation for 60 seconds to abandon the current timer and exit the set timer.

2.13.4 When the weekly timer is enabled, set the weekly timer at the current time and press and hold the " " key for 3 seconds to cancel the timer for the day.

2.13.5 **Timed display:** After setting the timing, the clock immediately updates the current state, executes power on during the power-on time period, executes power off outside the timing time period, and then executes once at the time. The " " and "OFF" icons are displayed when it is in the shutdown period, and the " " and the current working period "1/2/3" and "ON" are displayed when it is in the power-on period.

2.14 **ECO Mode Setting**

For TB series models, long press "△" or "▽" key in the main interface 3S to turn on/off ECO function, " " lights up when it is on.

2.15 **Timed return water temperature setting**

Long press " " key + "▽" key under the main interface for 3 seconds, you can enable or cancel the timer back to the water function mode, enable the timer back to the water function mode into the timer back to the water time settings, then " " and " " characters flashing at the same time, set the same method and set the switch timing

A total of three timer back to the water period can be set.

2.16 **Set timer refill function and (valid for commercial hot water models)**

Long press " " key + "△" key for 3 seconds under the main interface, you can enable or cancel the timer refill function mode, enable the timer refill function mode into the timer refill time settings, at this time " " and " " characters flashing at the same time, set the same method and set the switch timing.

A total of two timer back to the water period can be set

2.17 **Maintenance Mode**

Press and hold the "▽" key for more than 3 seconds to enter the maintenance mode, at this time the maintenance symbol " " is lit up and displayed. In this mode, you can check the information, controller self-test and other operations, press and immediately release the " " key or no key operation within 60 seconds will automatically return to the normal display state.

After entering the maintenance mode, the last viewed data code and its corresponding value are

displayed, and thereafter each time the "▽" key is pressed and immediately released or "△" key can be displayed in sequence as shown in the attached table.

Maintenance mode, press and release the " Ⓜ "key can display the main control board product information (temperature display area shows "r10" on behalf of the motherboard software version V1.1), the clock display area shows: "SBP2 "SBP2" represents the commercial inverter two-unit supply, "SBP3" represents the commercial inverter three-unit supply, "JdP" represents the home fixed-frequency machine, "SdP" represents commercial fixed-frequency machine; after pressing and releasing the " " key can display the product information of the line controller ("d10" represents software version V1.1), and return to the data display after 3 seconds.

Enter maintenance mode in the off state.

Press and hold the " " key for more than 5 seconds to restore the factory settings and exit.

Press and hold the " " key for more than 5 seconds to display self-test, all the display fields in turn after the end of the display self-test; display self-test after the end of the clock chip and memory chip, WIFI module detection. The left side of the temperature display area shows the detection result of the clock chip, the successful detection shows "OK", the failure shows "--", the right side shows the detection result of the memory chip, the successful detection shows "OK ", failure display "--". Clock area shows WIFi detection results, successful detection shows the current WIFI signal strength, failure shows "----", 3S after the end of the self-test

Press and hold the " Ⓜ " key for more than 5 seconds for output self-test, the relays of the control board are sucked and disconnected in turn, then end the output self-test.

Press and hold the " △ " key for more than 5 seconds to enter or exit the refrigerant charging or recovery mode, the "frost" character flashes in this mode, while the compressor, four-way valve, evaporator fan high speed, circulation pump and bypass valve are working, press any key or 20 minutes to exit automatically. After pressing any key or 20 minutes, it will exit automatically.

Note that the output self-test function is only for the quick test of the product, and it is strictly forbidden to use this function when it is under load. In order to avoid damage to the unit by misoperation, the factory setting cannot be restored after the controller is powered on for 5 minutes, and the output self-test and refrigerant charging or recovery mode cannot be entered.

In maintenance mode, press and hold the " "+" +" △"+" ▽ "key for 5 seconds to enter the model parameter setting Work-installation mode, which is shown on the right.

In the work mode, press and hold the " "+ "+" △"+" ▽ "key for 5 seconds, then exit the work mode and carry out normal work display.

Work mode, power on and after the initialization is complete, the display board cycle 1 second to send a setting command, and according to the motherboard to reply the results of success or failure instructions. Among them, when the setting is successful, "JC" and "0" are always displayed; when the setting fails, "JC" is always displayed and "0 " flashes.

In the tooling mode, the model number "0" can be adjusted by " △" key and " ▽ " key, and in the process of model adjustment, "JC " and "0" are both flashing, you can confirm the setting by " Ⓜ " key

2.18 **Online fast detection mode (valid for commercial machines with fixed frequency)**

Power on and within 5 minutes, press and hold the " "+" Ⓜ "+"▽ "keys simultaneously for more than 5 seconds to enable the fast detection mode, which optimizes the press start-up delay, as follows :

(1) The time to keep the electronic expansion valve initialized for 60 seconds before the press starts is shortened to 3 seconds.

(2) The default 90-second time for the circulating water pump to start the press in advance is shortened to 15 seconds.

(3) The electronic expansion valve opens up to 480 steps after the press stops and maintains a 2-minute delay time cancellation.

In the fast detection mode, press and hold " "+" Ⓜ "+" ▽ "at the same time for more than 5 seconds, then the fast detection mode is cancelled and normal control is restored. Power

off and power on again, then normal control is automatically restored.

2.19 **Usage period password setting**

In the unlocked state while holding down 5 keys 5S until "beep" into the use of the term password input, then the temperature zone display password "0000", you can press "△" or "▽" key to enter the password, and then press "M" to switch to the next password input, enter the 4-bit password and press "M" key to confirm, the password is correct to enter the term settings, then the clock area is displayed before the set value, press "△" or "▽" key to adjust, and then press "M" to confirm, return to the main interface, press and immediately release the " ⏻ "key or no key operation within 60 seconds will automatically return to the normal display state, does not save the set value.

2.20 **Photovoltaic timing (valid for home machines)**

Under the main interface long press the " ⏻ "+" 🕒 "+" Ⓜ "key for 3 seconds to enter the PV timing settings, press and immediately release " 🕒 " key can switch the hour part, minute part, hour part and minute part of the end time of the two time periods in turn, flashing display when switching to the corresponding value, while displaying the corresponding time period and flashing display "ON" or "OFF " symbol, press " △ " or " ▽ " can be adjusted up or down and flashing display the corresponding value. After setting the timing period, press and release the " ⏻ " key immediately or no key operation within 15 seconds can save the modification and return to normal display status.

2.21 **Networking Features**

Network function display: when there is a slave access to the host, the host line controller display " 🖧 " indicates that the device uses multiple devices networking function. Due to the display area limitation, the number shown in " ①②③④⑤⑥⑦⑧ " during the operation of the equipment indicates that the N (1~8) slave accesses the network connected to the host, and the actual network can be grouped 16 units (1 master, 15 slaves).

How to query the operating parameters of master and slave machines under the network function: long press " △ " for 5 seconds to enter the query of master and slave machine parameters, the symbol " 8 " indicates the master and slave machine number (0 is the master, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E F represents the slave code), switch the number of the group being checked by tapping " 🕒 " key. After entering a host or slave, tap " △ ", " ▽ " to scroll through the lines to display each "operation parameter"; press "On/Off " key or no key operation for 60 seconds to exit automatically to view the parameter status.